

# Ethanol Blending in Gasoline – Vietnam

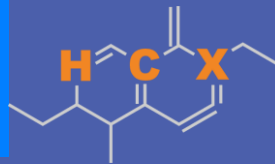
October 2025



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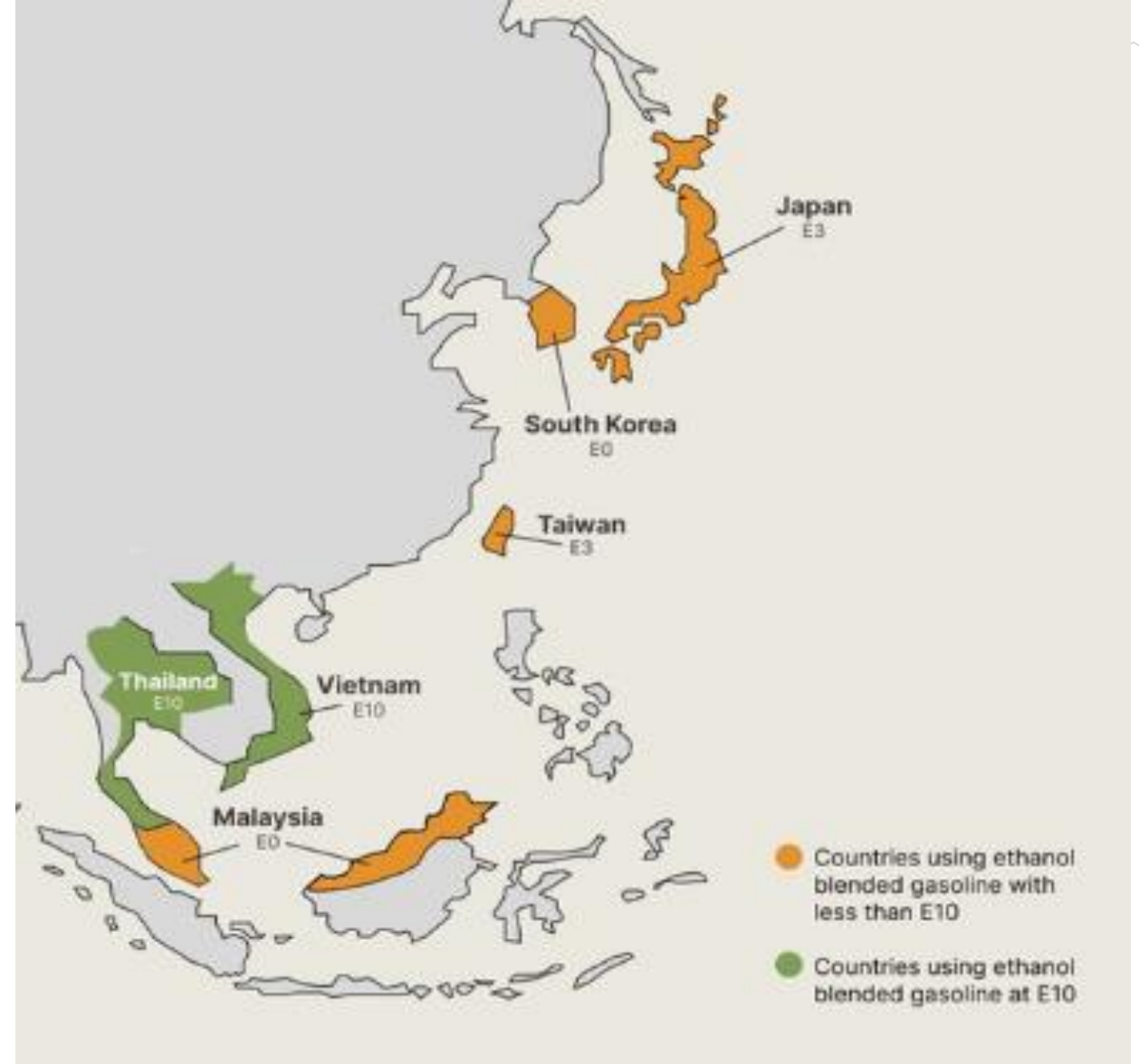


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# Ethanol Blending in Asia

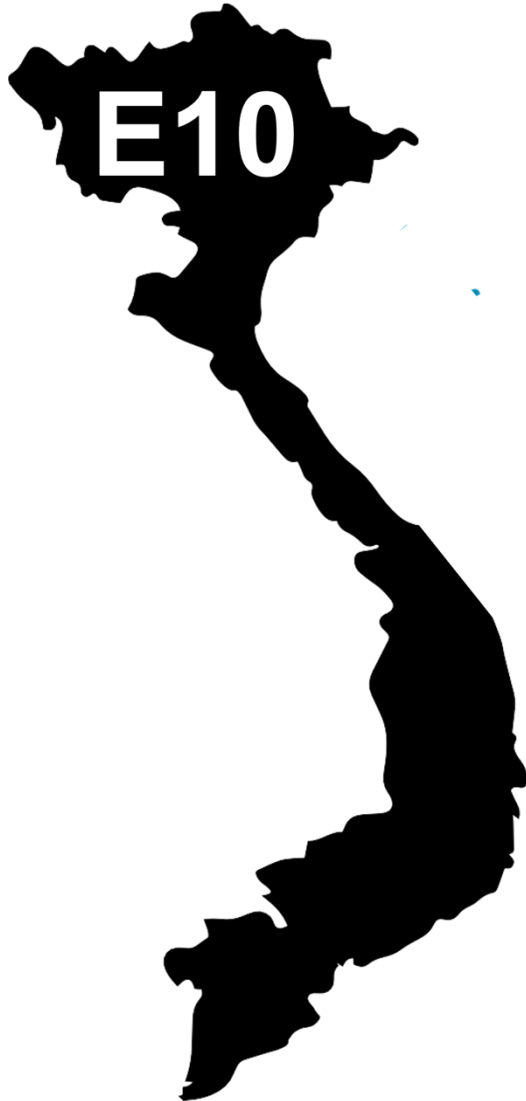
There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

# Ethanol Gasoline Blending in Vietnam

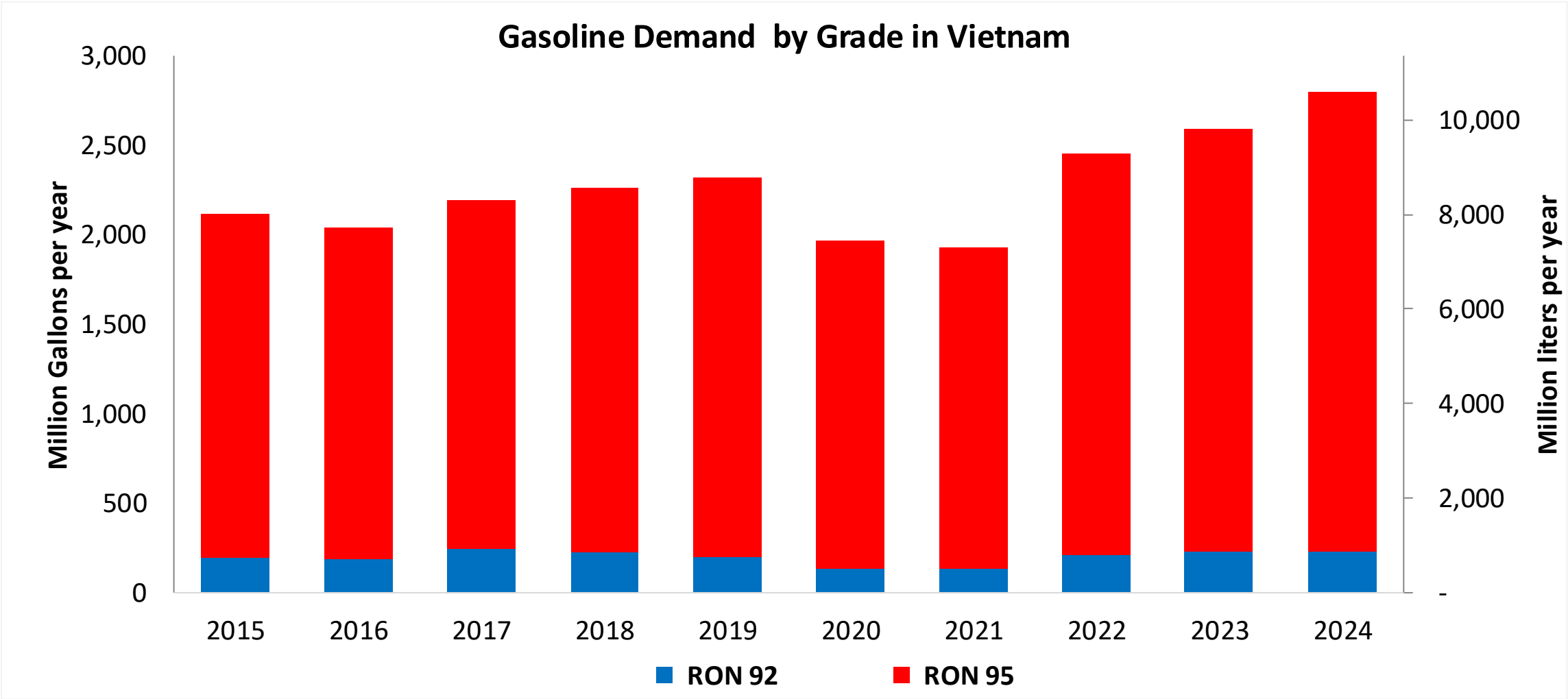


In 2024, gasoline consumption in Vietnam was 9,384 million liters (2,479 million gallons) and growth rate was 1.0%. Gasoline production and net imports were 7,870 and 1,514 million liters (2,079 and 400 million gallons) correspondingly. Vietnam complies offers three grades: RON 92, RON 95 and RON 98 with an average octane of 94.8 RON.

Vietnam's bioethanol demand is growing rapidly, driven by government policies like a planned nationwide E10 gasoline mandate set to begin in 2026. Ethanol consumption in Vietnam was 236 million liters (62 million gallons) representing 2.5% of gasoline demand. Ethanol production and Net Imports were 200 and 36 million liters (62 and 9 million gallons) correspondingly.

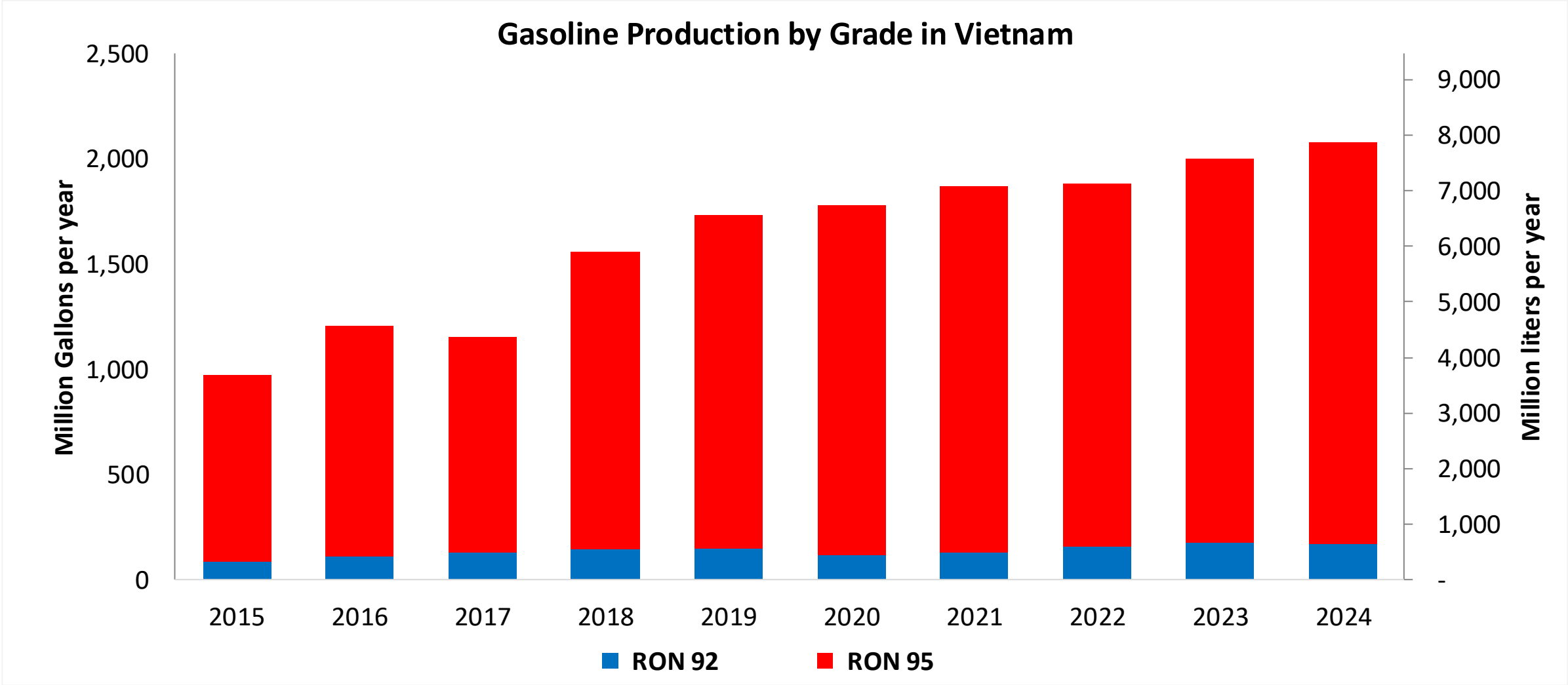
Source: Petro Vietnam, EIA,, EITdata, USDA

# Gasoline Demand in Vietnam



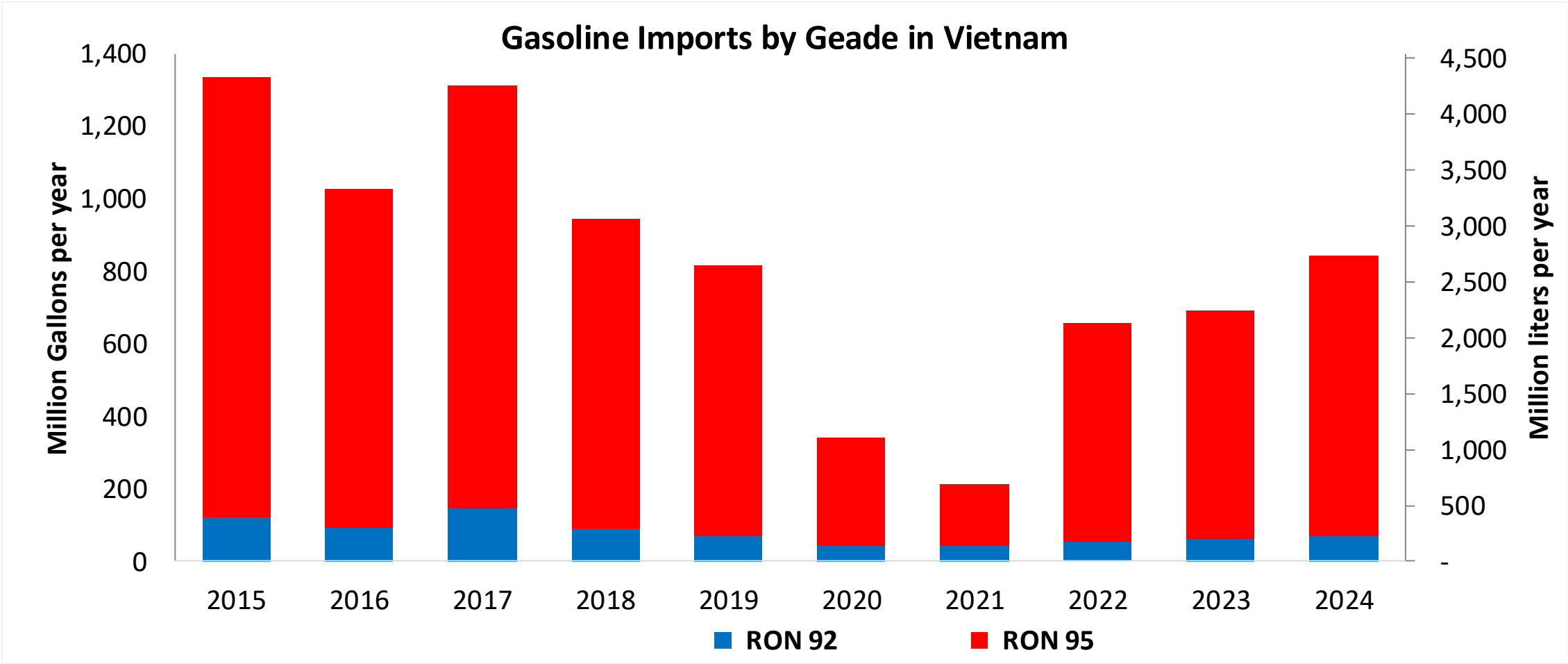
Source: Petro Vietnam, EIA,, EITdata

# Gasoline Production in Vietnam



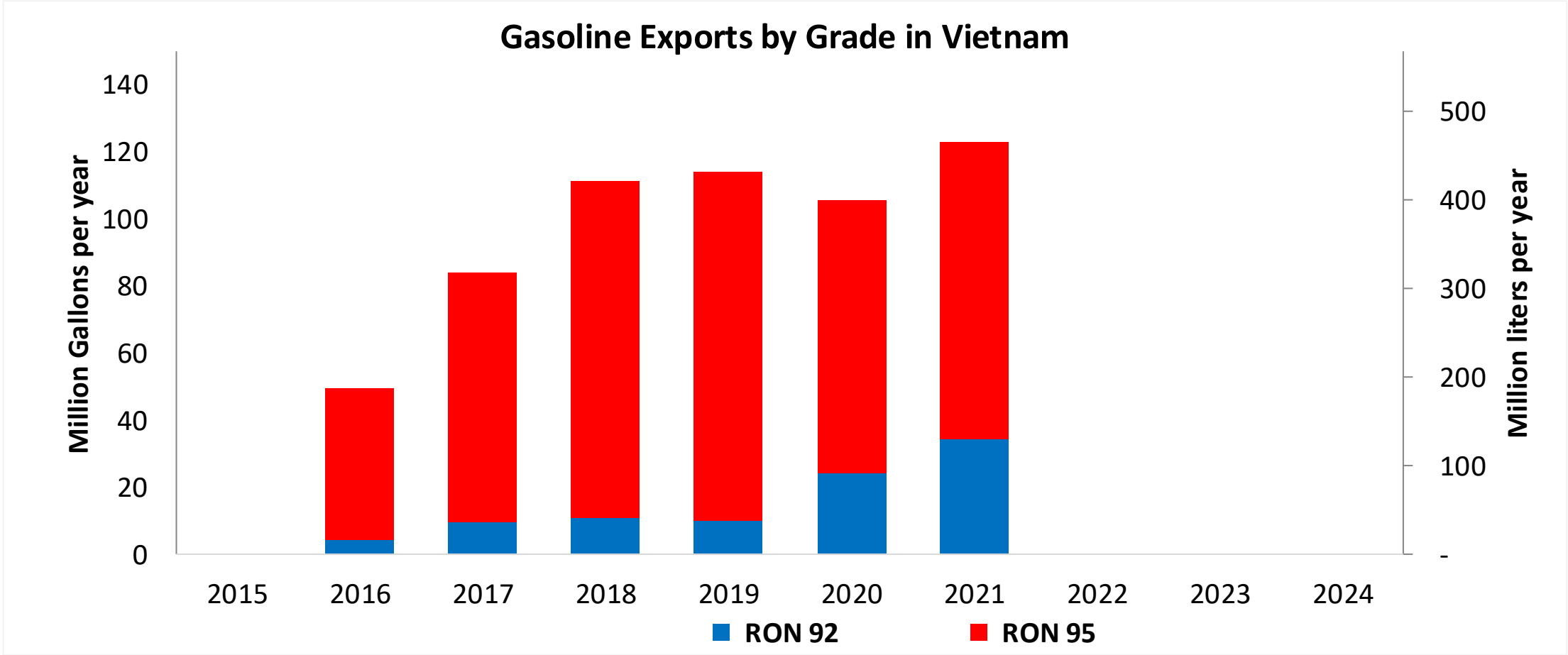
Source: Petro Vietnam, EIA,, EITdata

# Gasoline Imports in Vietnam



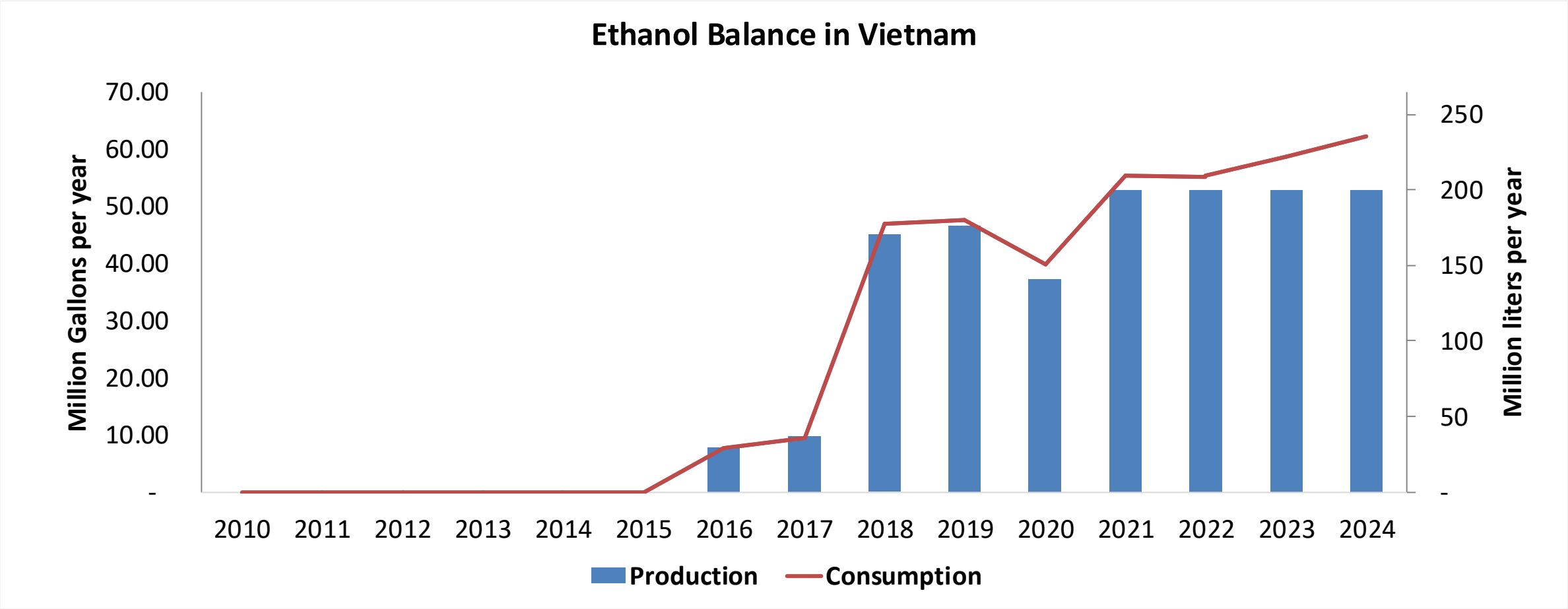
Source: Petro Vietnam, EIA,, EITdata

# Gasoline Exports in Vietnam



Source: Petro Vietnam, EIA,, EITdata

# Ethanol Balance in Vietnam



Source: Petro Vietnam, USDA

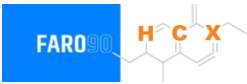


# Gasoline Quality in Vietnam

| Name                | QCVN 01:2022/BKHCN |       |        |       |        |       |
|---------------------|--------------------|-------|--------|-------|--------|-------|
| Implementation Year | 2023               |       |        |       |        |       |
| Applicability       | All territory      |       |        |       |        |       |
| Grade               | RON 92             |       | RON 95 |       | RON 97 |       |
|                     | Min                | Max   | Min    | Max   | Min    | Max   |
| RON                 | 92.0               |       | 95.0   |       | 97.0   |       |
| MON                 | -                  |       | -      |       | -      |       |
| AKI                 | -                  |       | -      |       | -      |       |
| Benzene (%vol)      |                    | 1.0   |        | 1.0   |        | 1.0   |
| Aromatics (% vol)   |                    | 35.0  |        | 35.0  |        | 35.0  |
| Olefins (%vol)      |                    | 18.0  |        | 18.0  |        | 18.0  |
| Lead (g/L)          |                    | 0.005 |        | 0.005 |        | 0.005 |
| Manganese (mg/L)    |                    | 5.0   |        | 5.0   |        | 5.0   |
| Sulfur (ppm)        |                    | 10    |        | 10    |        | 10    |
| Oxygen (%w)         |                    | 2.7   |        | 2.7   |        | 2.7   |
| Ethanol (%vol)      |                    | -     |        | -     |        | -     |
| MTBE (%vol)         |                    | -     |        | -     |        | -     |
| RVP (psi)           |                    | 8.7   |        | 8.7   |        | 8.7   |

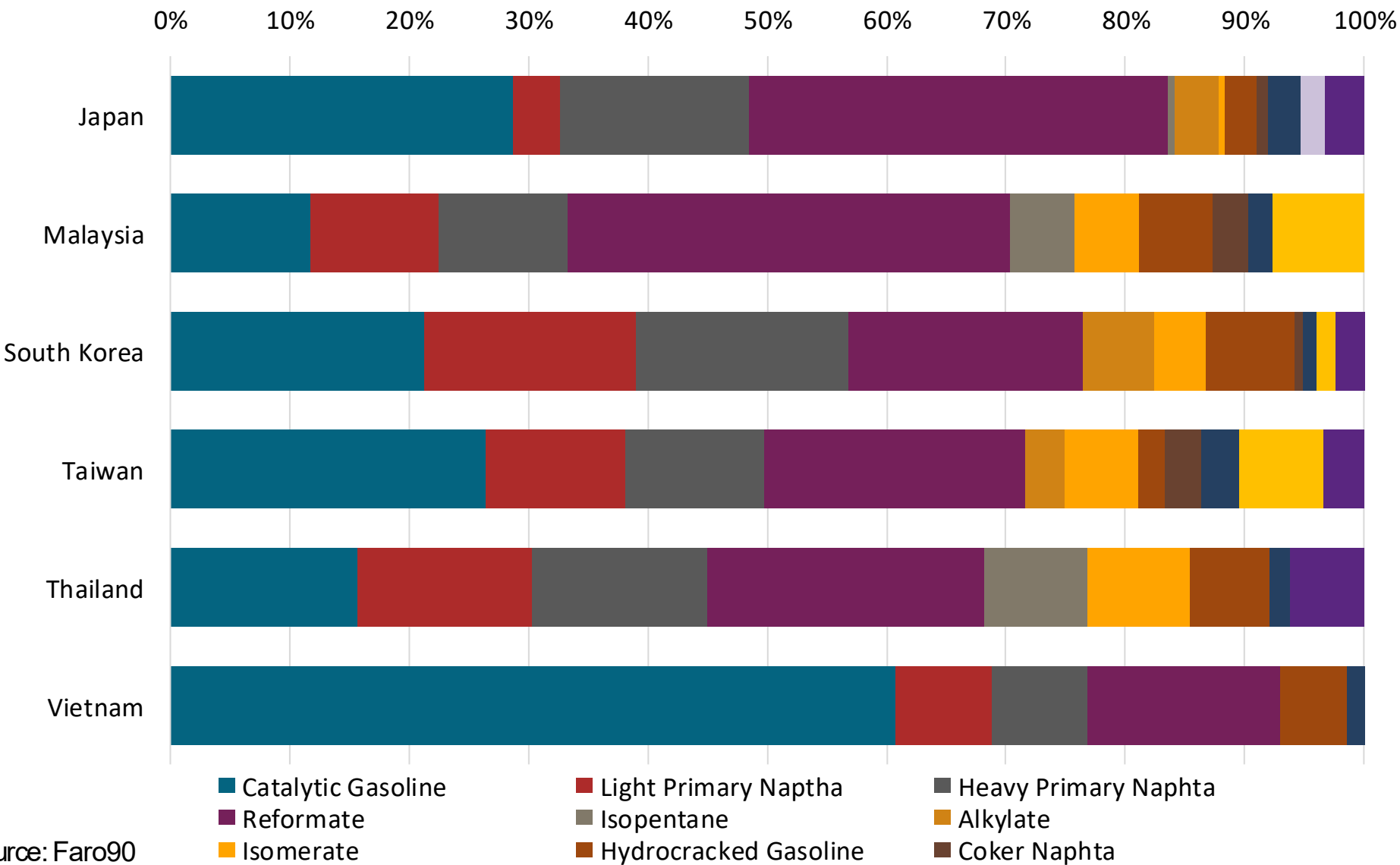
Source: Caselaw

# Gasoline Component Blending in Asia



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

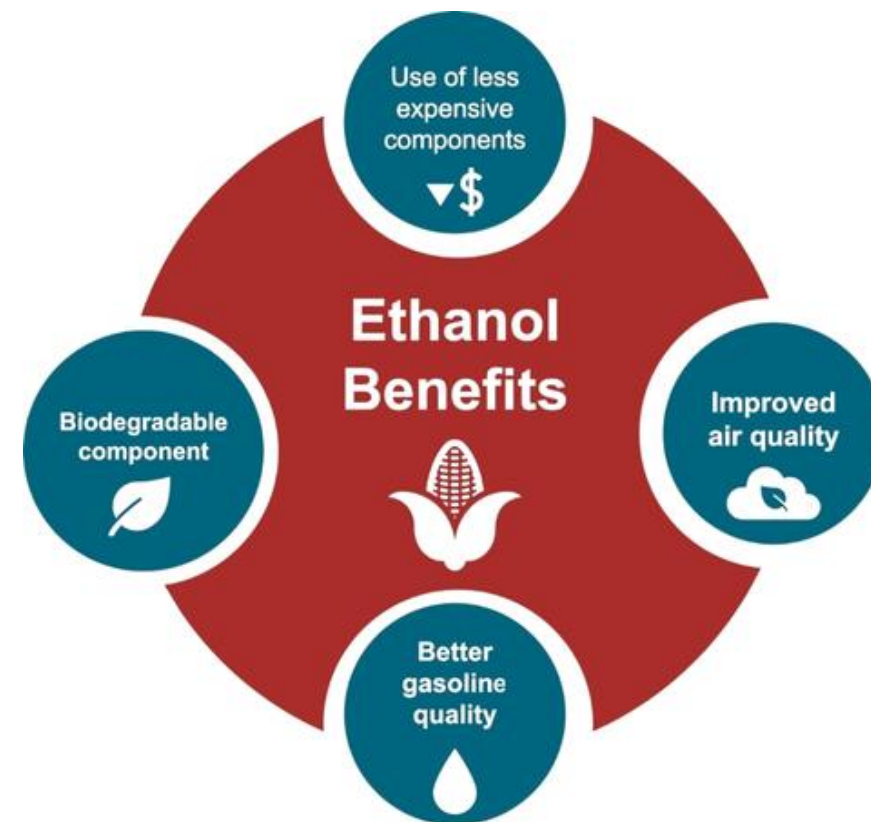
# Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

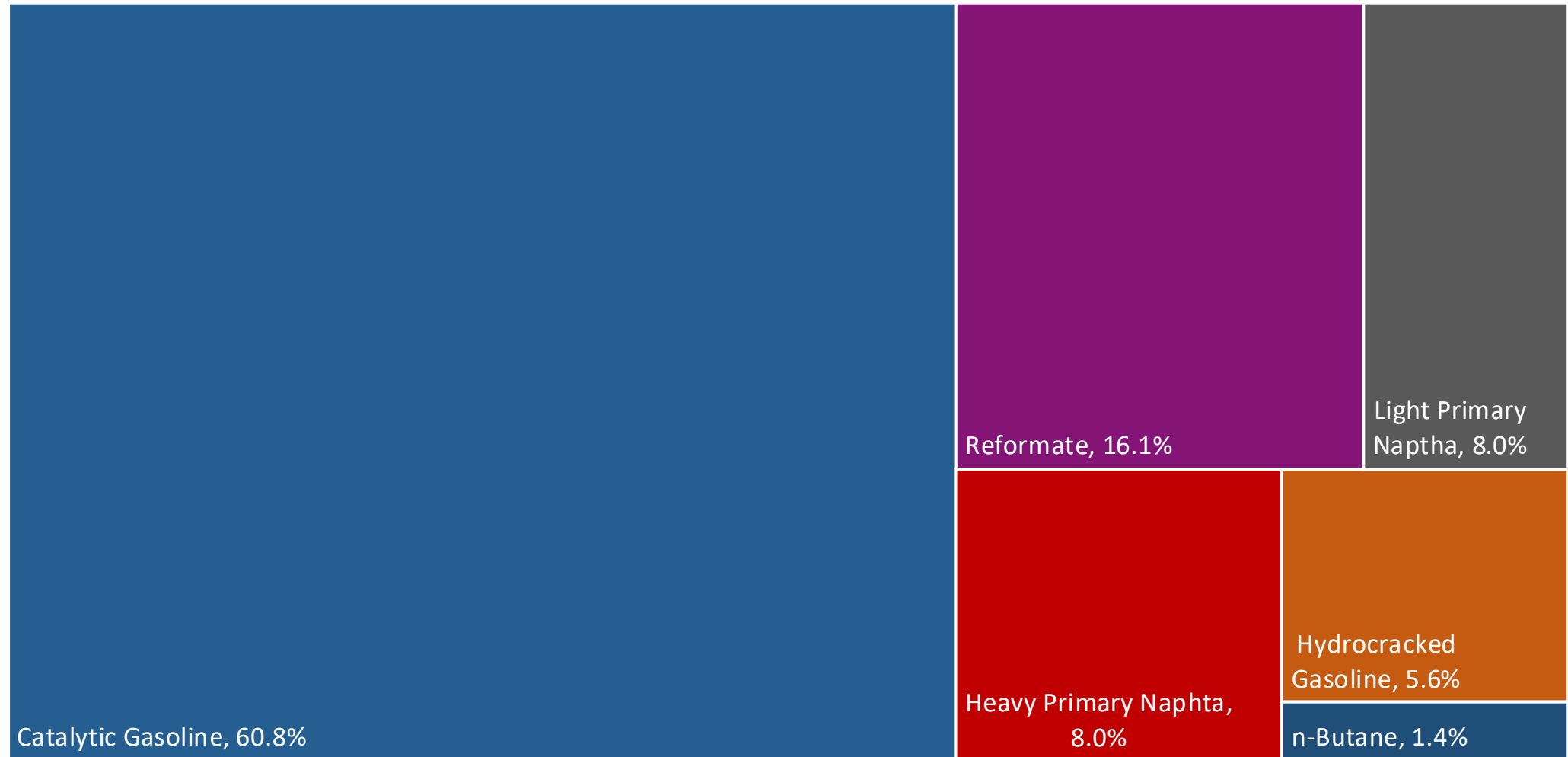
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

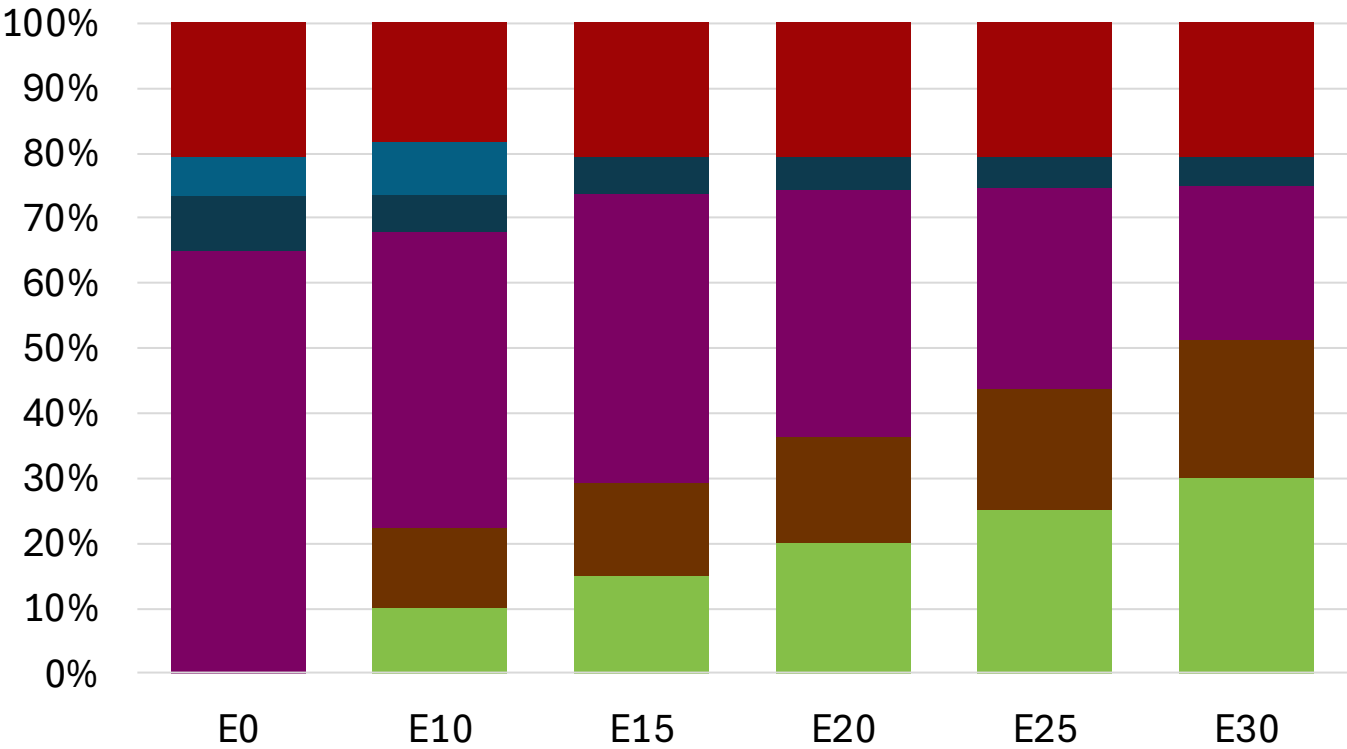


# Vietnam Available Blending Components



Source: Faro90

# Vietnam – Gasoline 92 – Constant Octane

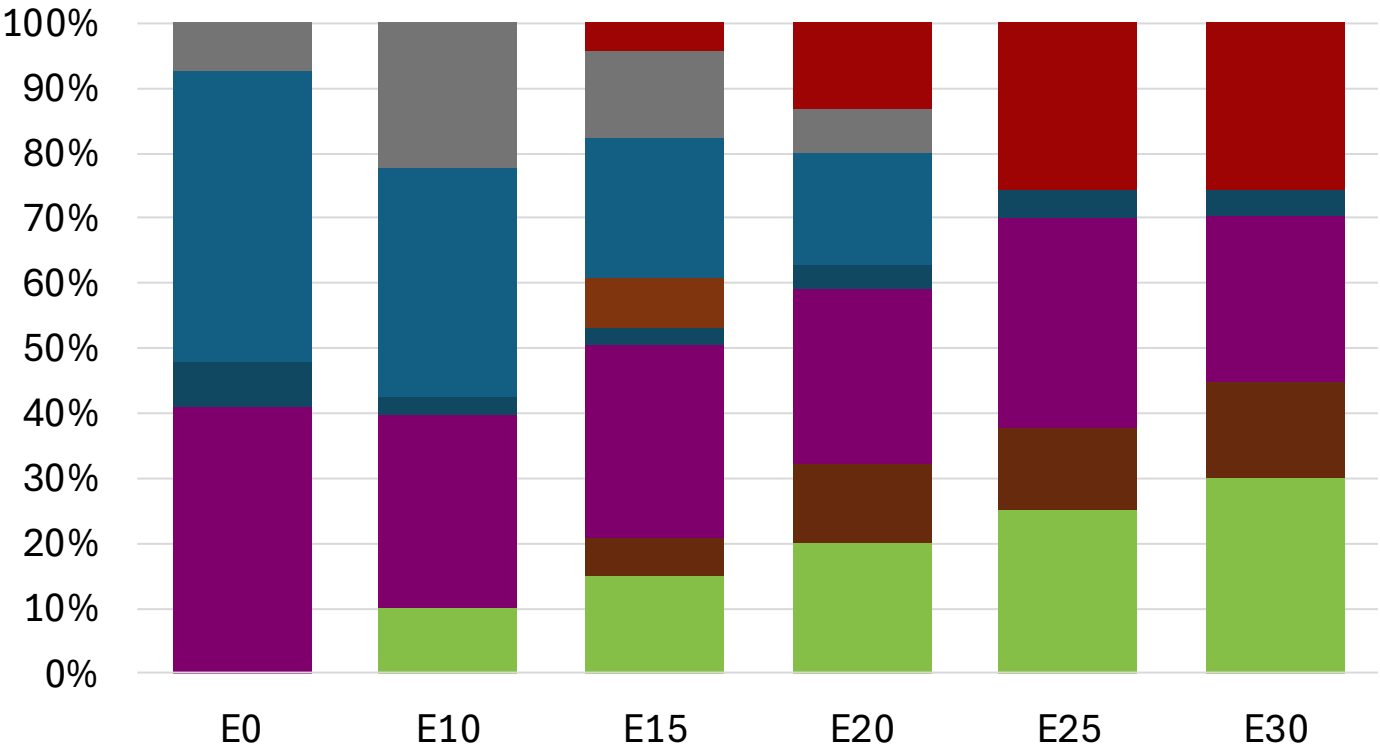


Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 92.0     | 92.0     | 92.9     | 94.0     | 95.0     | 96.0     |
| Price (US\$/gal) | \$ 2.280 | \$ 2.113 | \$ 2.063 | \$ 2.020 | \$ 1.972 | \$ 1.921 |

Source: Faro90

# Vietnam – Gasoline 95 – Constant Octane

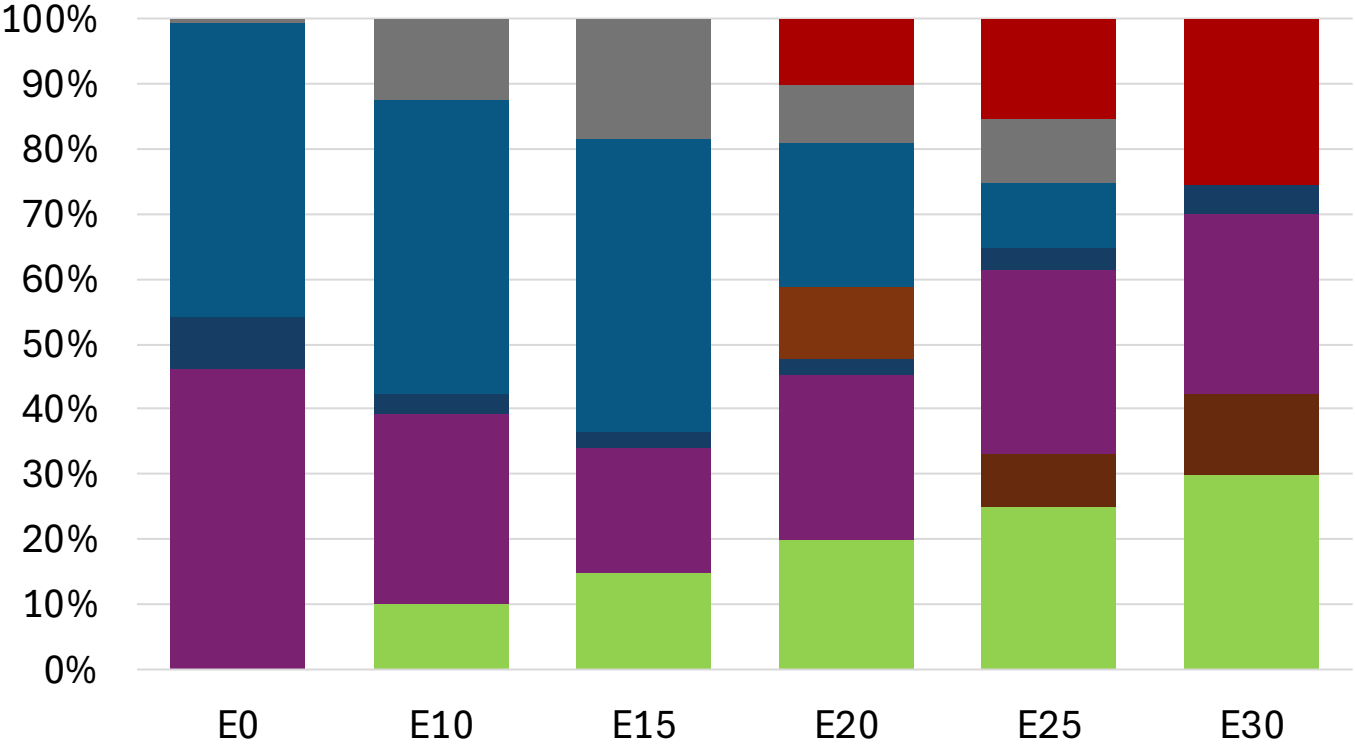
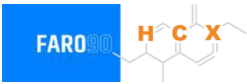


Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 95.0     | 95.0     | 95.0     | 95.0     | 95.2     | 96.3     |
| Price (US\$/gal) | \$ 2.436 | \$ 2.305 | \$ 2.214 | \$ 2.090 | \$ 1.981 | \$ 1.933 |

Source: Faro90

# Vietnam - Gasoline 97 - Constant Octane

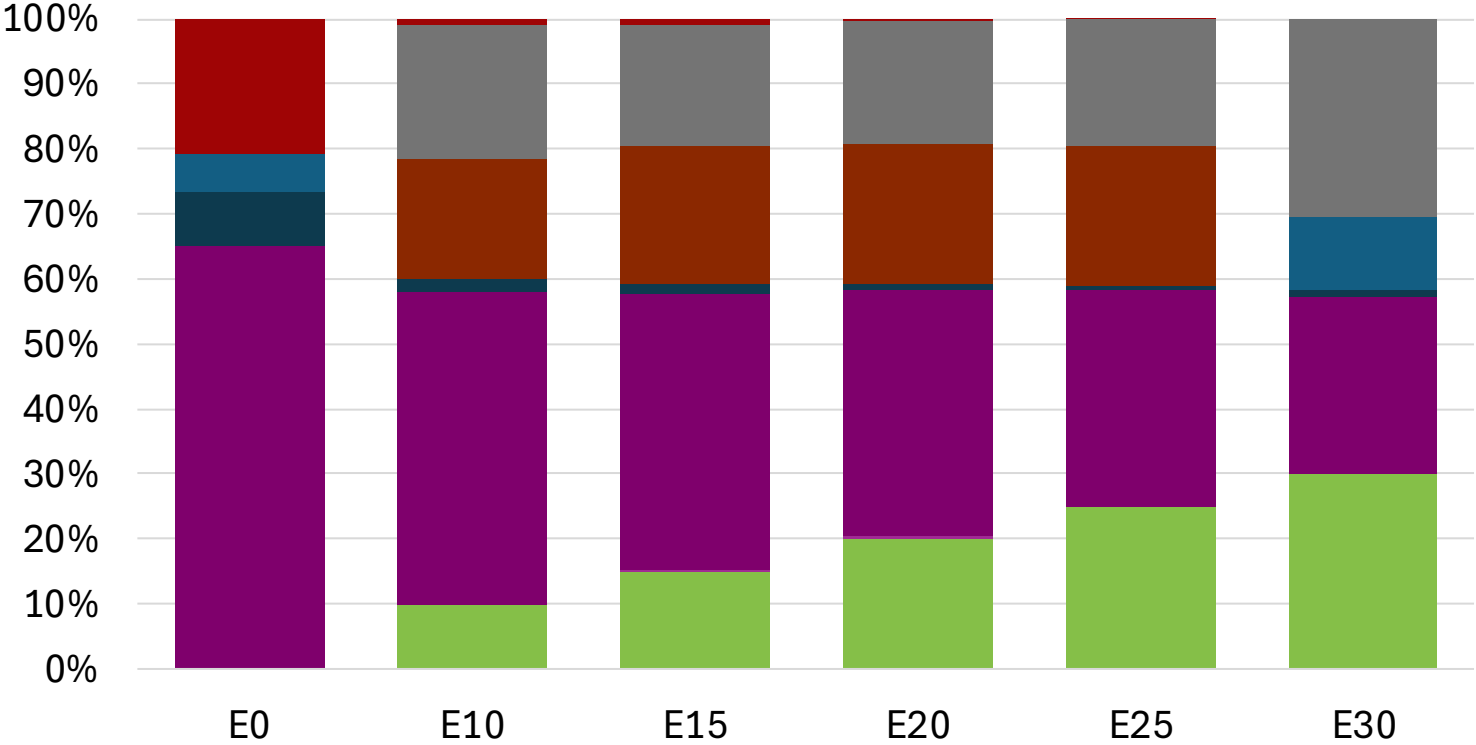


Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 97.0     | 97.0     | 97.0     | 97.0     | 97.0     | 97.0     |
| Price (US\$/gal) | \$ 2.542 | \$ 2.416 | \$ 2.322 | \$ 2.228 | \$ 2.105 | \$ 1.983 |

Source: Faro90

# Vietnam – Gasoline 92 – Increased Octane



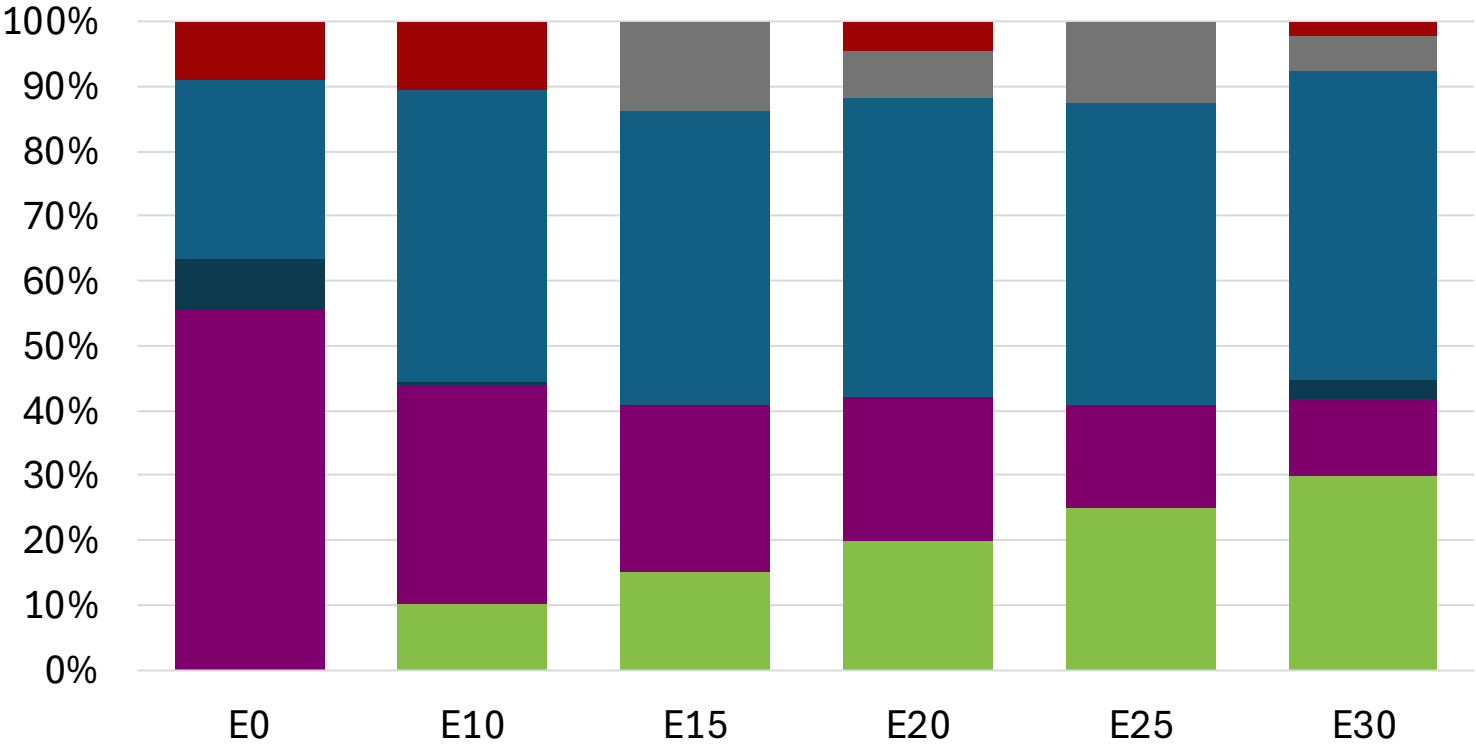
Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 92.0     | 94.1     | 95.7     | 97.4     | 99.0     | 101.0    |
| Price (US\$/gal) | \$ 2.280 | \$ 2.280 | \$ 2.280 | \$ 2.280 | \$ 2.280 | \$ 2.280 |

Source: Faro90



# Vietnam – Gasoline 95 – Increased Octane

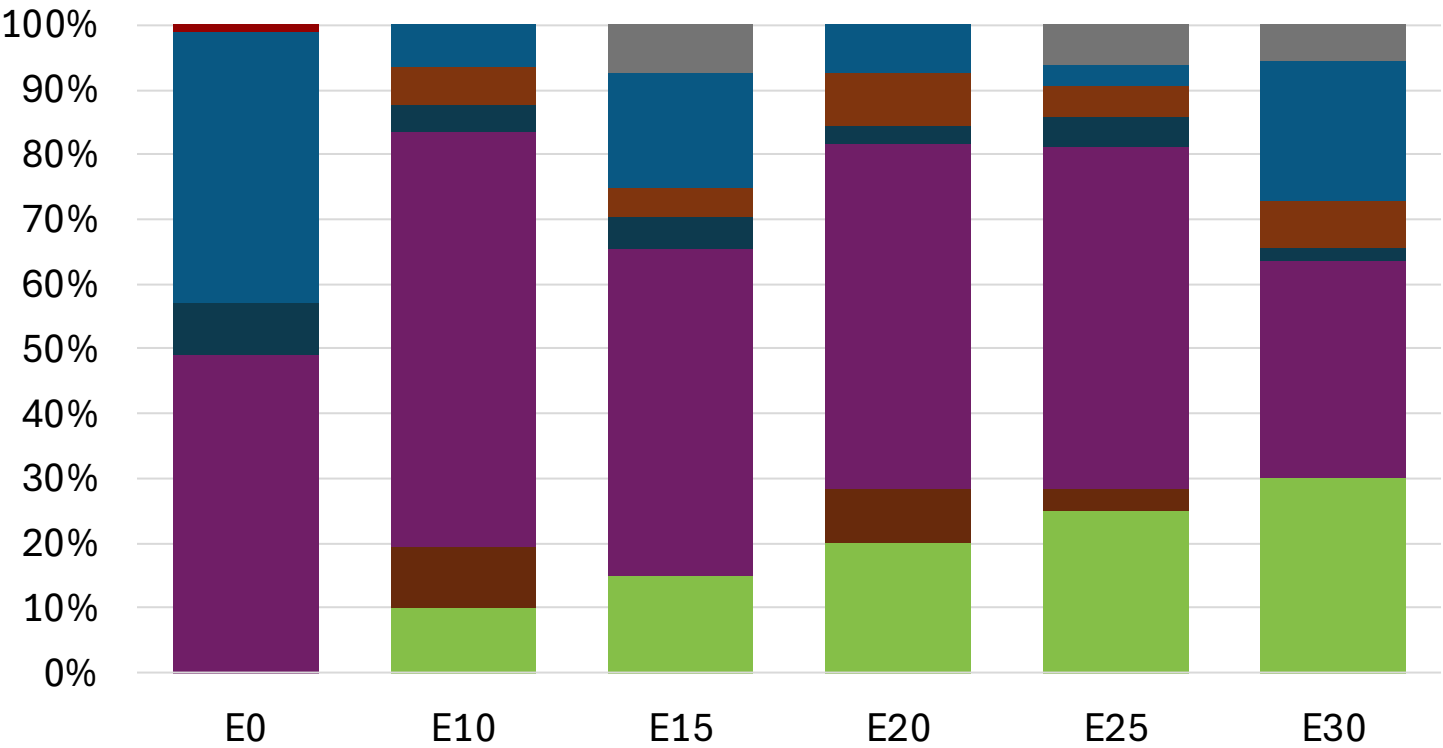


Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 95.0     | 96.8     | 98.4     | 100.0    | 101.6    | 103.7    |
| Price (US\$/gal) | \$ 2.436 | \$ 2.436 | \$ 2.436 | \$ 2.436 | \$ 2.436 | \$ 2.436 |

Source: Faro90

# Vietnam – Gasoline 97 – Increased Octane



Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 97.0     | 99.1     | 101.1    | 101.9    | 103.9    | 104.9    |
| Price (US\$/gal) | \$ 2.541 | \$ 2.541 | \$ 2.541 | \$ 2.541 | \$ 2.541 | \$ 2.541 |

Source: Faro90

# Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

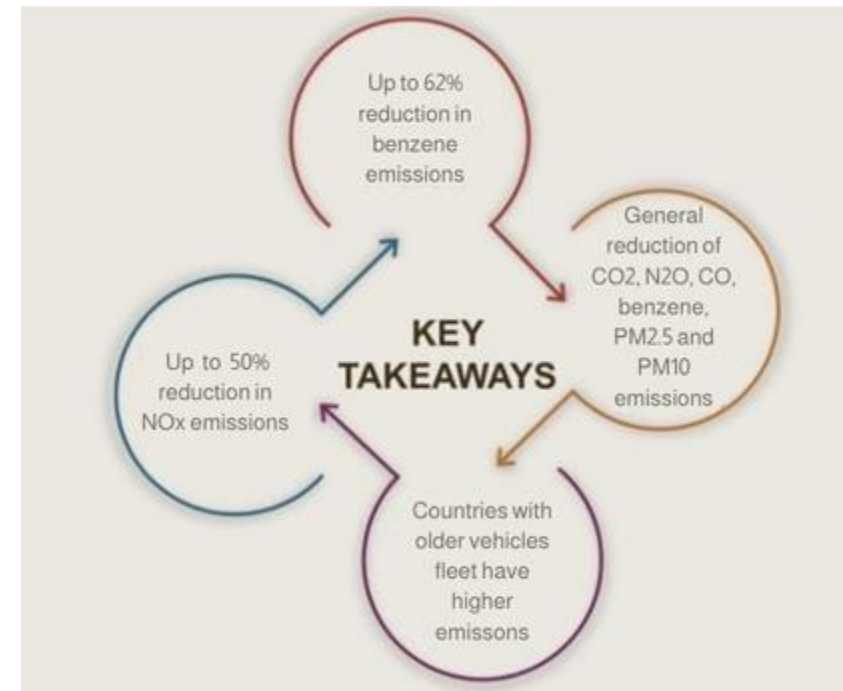
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

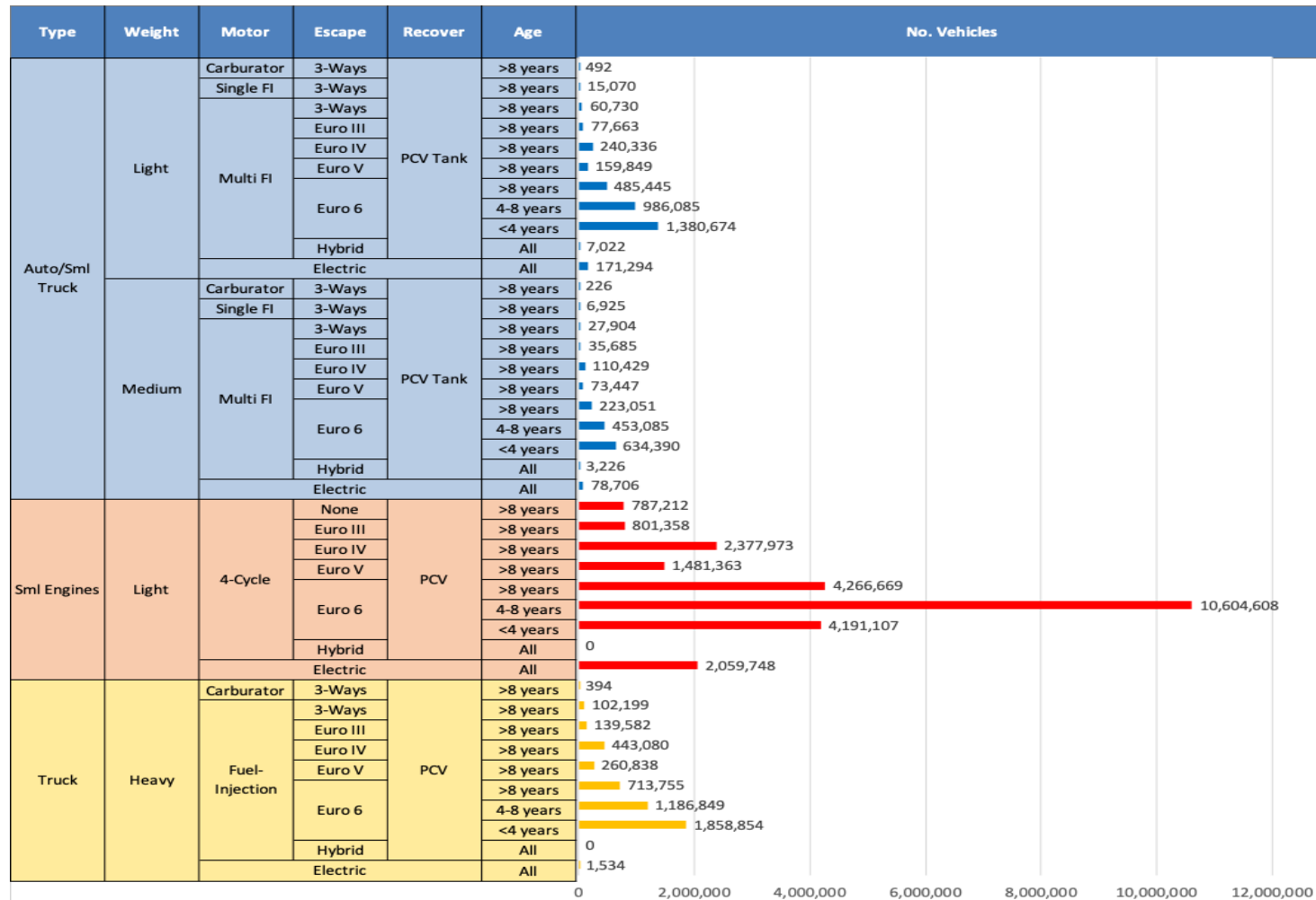
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet\*.

*\*Source: Bureau of transportation statistics.*

## Main Results



# Gasoline Vehicle Fleet - Vietnam

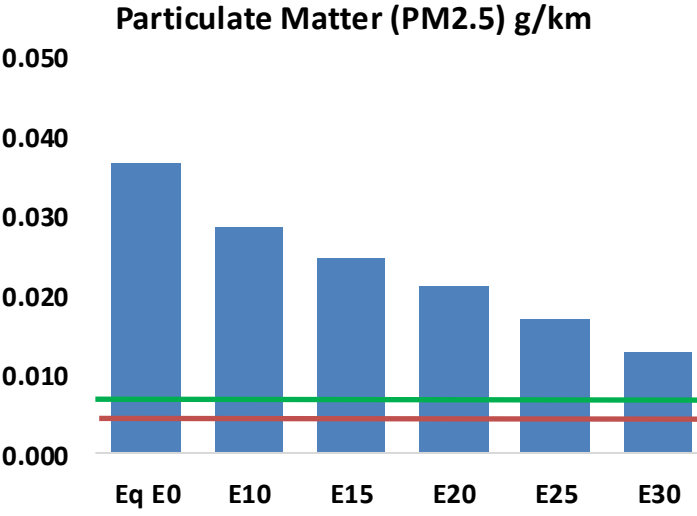
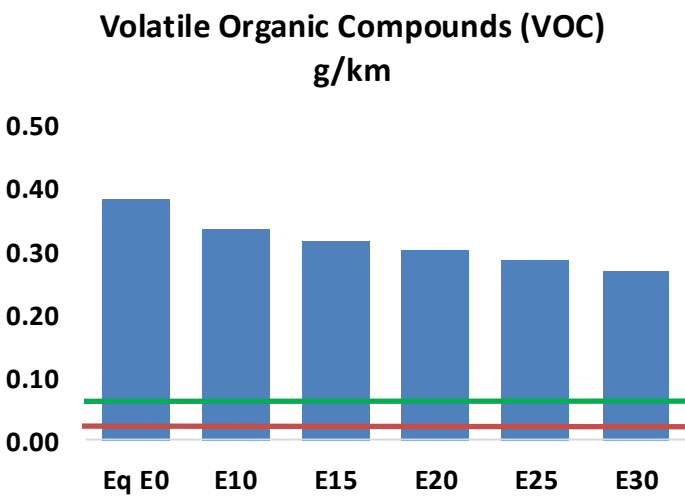
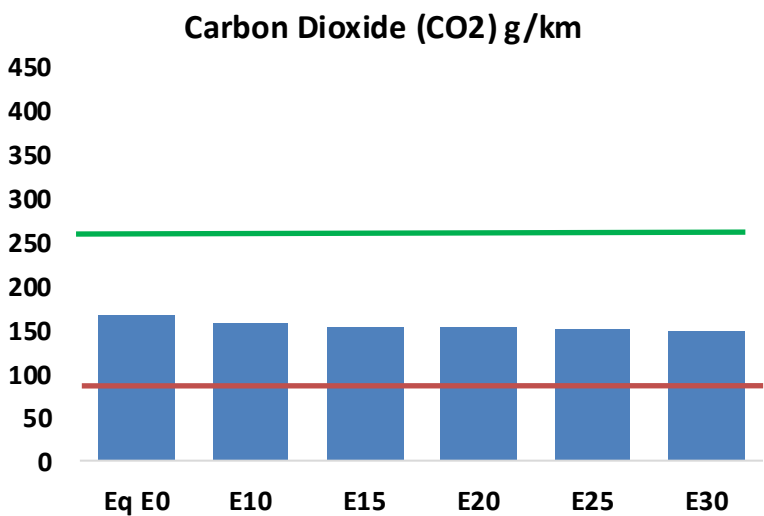
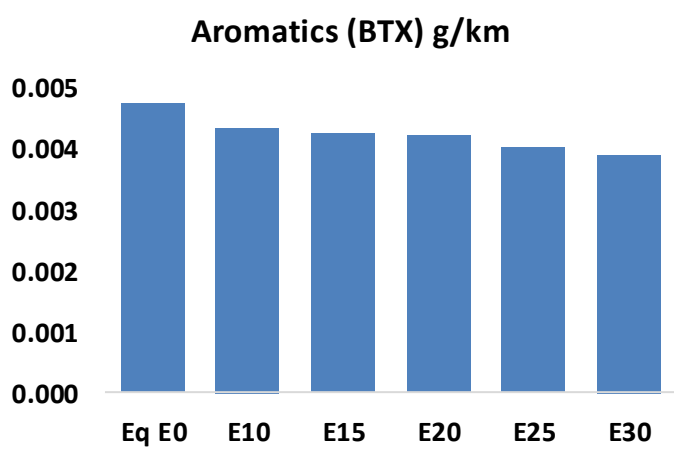
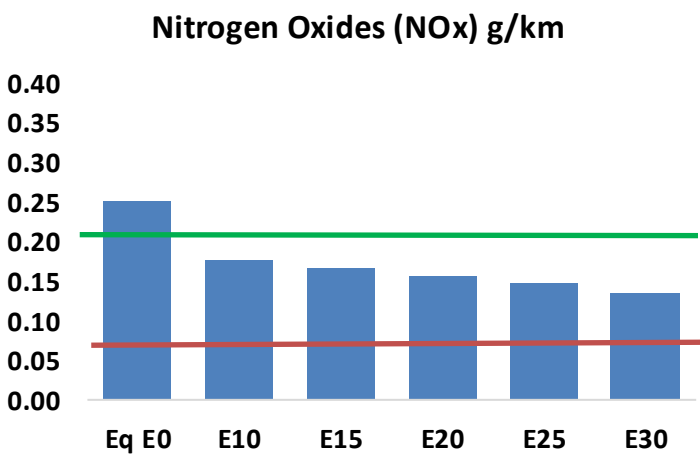
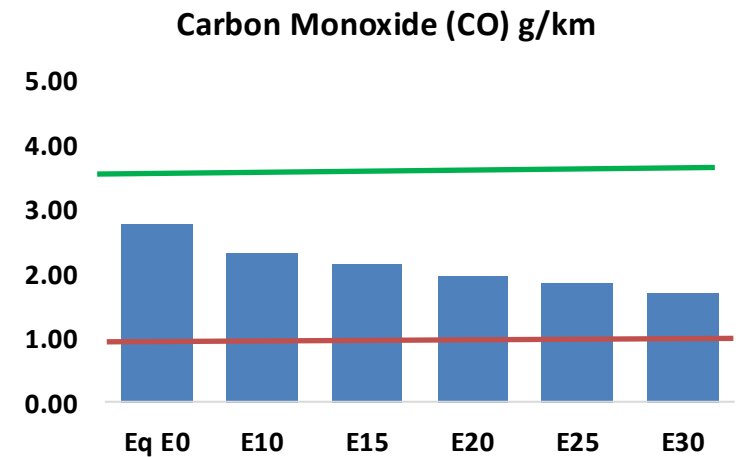
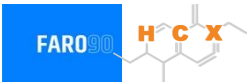


Vehicle Fleet: 35,508,859  
Gasoline Vehicle Fleet: 25,550,917

Source: MoIT

# Vietnam – Vehicular Emissions

TIER III BIN 125 United States  
Euro 6 Standard



# Vietnam – Vehicular Emissions

| Vehicular Fleet Emissions (kg/day) | Eq E0      | E5         | E10        | E15        | E20        | E25        | E30        | Reduction (%) |      |
|------------------------------------|------------|------------|------------|------------|------------|------------|------------|---------------|------|
|                                    |            |            |            |            |            |            |            | E10           | E20  |
| <b>CO</b>                          | 1,359,820  | 1,252,428  | 1,145,036  | 1,056,476  | 967,099    | 898,296    | 825,989    | -16%          | -29% |
| <b>CO2</b>                         | 80,379,150 | 78,312,257 | 76,360,192 | 74,824,950 | 74,065,253 | 73,754,180 | 72,394,679 | -5%           | -8%  |
| <b>VOC</b>                         | 186,910    | 175,504    | 164,098    | 155,329    | 146,771    | 140,271    | 131,313    | -12%          | -21% |
| <b>NOx</b>                         | 122,884    | 92,163     | 86,019     | 81,073     | 76,459     | 71,371     | 65,989     | -30%          | -38% |
| <b>SOx</b>                         | 872        | 806        | 740        | 685        | 631        | 573        | 512        | -15%          | -28% |
| <b>NH3</b>                         | 33,331     | 33,001     | 32,671     | 32,826     | 33,195     | 33,455     | 33,671     | -2%           | 0%   |
| <b>N2O</b>                         | 1,350      | 1,342      | 1,337      | 1,343      | 1,371      | 1,386      | 1,402      | -1%           | 2%   |
| <b>CH4</b>                         | 47,919     | 47,919     | 47,919     | 48,637     | 49,692     | 50,602     | 51,465     | 0%            | 4%   |
| <b>Lead</b>                        | -          | -          | -          | -          | -          | -          | -          | 0%            | 0%   |
| <b>Butadiene</b>                   | 1,465      | 1,391      | 1,318      | 1,265      | 1,216      | 1,179      | 1,126      | -10%          | -17% |
| <b>Acetaldehyde</b>                | 3,770      | 4,327      | 6,348      | 9,668      | 13,171     | 15,051     | 17,784     | 68%           | 249% |
| <b>Formaldehyde</b>                | 14,599     | 14,194     | 16,546     | 19,429     | 20,354     | 22,517     | 24,513     | 13%           | 39%  |
| <b>Benzene</b>                     | 2,328      | 2,237      | 2,120      | 2,086      | 2,069      | 1,979      | 1,915      | -9%           | -11% |
| <b>PM2.5</b>                       | 5,773      | 5,126      | 4,479      | 3,889      | 3,303      | 2,680      | 2,031      | -22%          | -43% |
| <b>PM10</b>                        | 12,169     | 10,805     | 9,442      | 8,199      | 6,963      | 5,650      | 4,281      | -22%          | -43% |

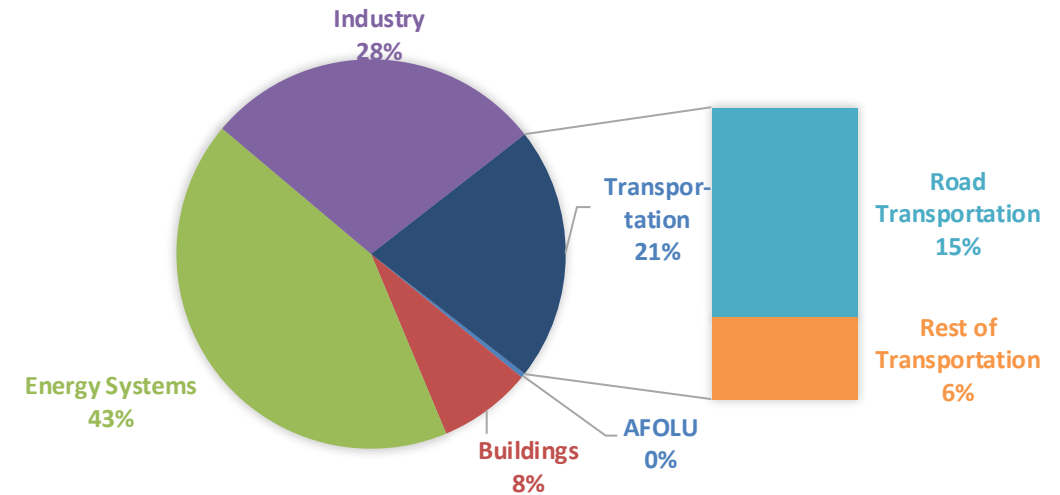
# Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

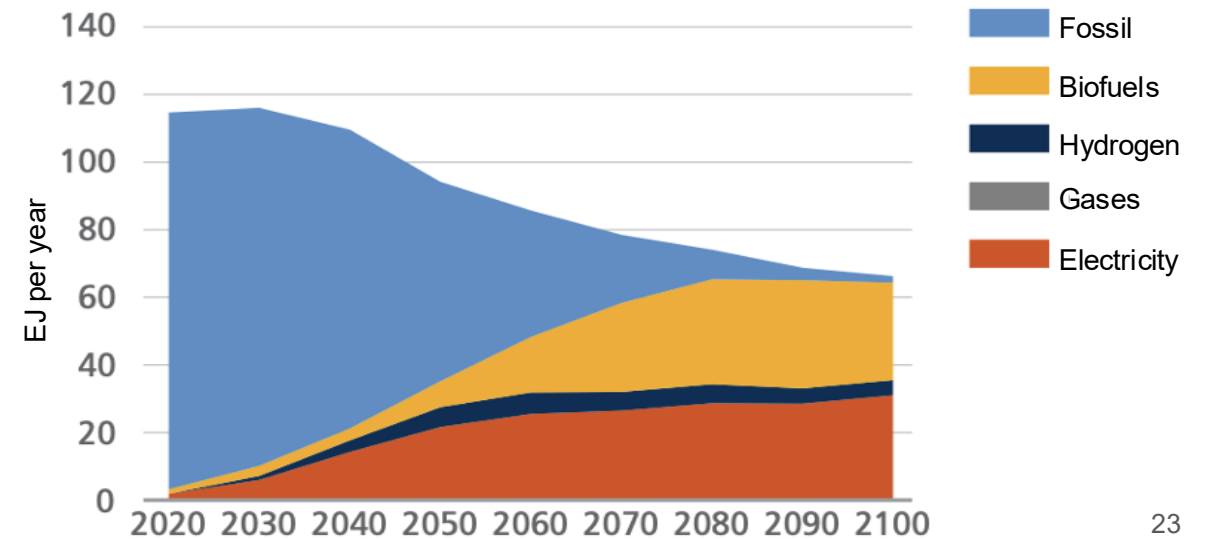
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

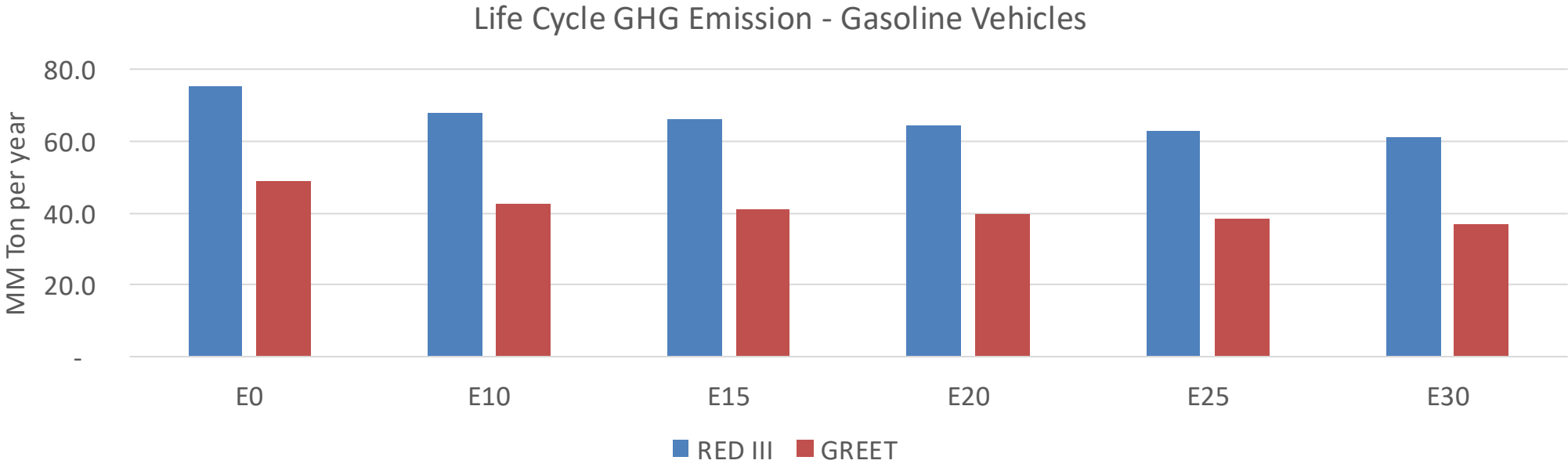
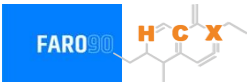
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



# Vietnam – Gasoline Vehicle Fleet Life Cycle GHG Emissions



|                                  |       |
|----------------------------------|-------|
| Country Emissions 2024 (MMT/año) | 489.9 |
| Target @ 2035 (MMT/year)         | 330.6 |
| Delta                            | 159.3 |

| %Reduction vs E0 | E0   | E10   | E15   | E20   | E25   | E30   |
|------------------|------|-------|-------|-------|-------|-------|
| REET 2024        | 0.0% | 13.3% | 16.6% | 19.3% | 21.8% | 25.2% |
| RED II/III       | 0.0% | 9.6%  | 12.2% | 14.3% | 16.3% | 18.9% |

| % Target@2035 | E0   | E10  | E15  | E20  | E25  | E30  |
|---------------|------|------|------|------|------|------|
| REET 2024     | 0.0% | 4.1% | 5.1% | 5.9% | 6.7% | 7.7% |
| RED II/III    | 0.0% | 4.5% | 5.7% | 6.7% | 7.7% | 8.9% |

Source: Greet, RED II/III, climateactiontracker.org, F90